Assignment 9

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Labwork Database

1. Create the MariaDB database **labwork** with 3 tables (customer/orders/salesman).

- CREATE DATABASE labwork;

- CREATE TABLE salesman (
  salesman_id INT PRIMARY KEY,
  name VARCHAR(100),
  city VARCHAR(100),
  commission DECIMAL(4, 2)
  );

- CREATE TABLE customer (
  customer_id INT PRIMARY KEY,
  customer_name VARCHAR(100),
  city VARCHAR(100), grade INT,
  salesman_id INT,
  FOREIGN KEY (salesman_id)
  REFERENCES salesman(salesman_id)
  );

- CREATE TABLE orders (
  order_no INT PRIMARY KEY,
  purchase_amt DECIMAL(10, 2),
  order_date DATE,
  customer_id INT,
  salesman_id INT,
  FOREIGN KEY (customer_id)
  REFERENCES customer(customer_id),
  FOREIGN KEY (salesman_id)
  REFERENCES salesman(salesman_id)
  );

2. Fill the tables with test data.

- INSERT INTO salesman (salesman_id, name, city, commission) VALUES
  (5001, 'James Hoog', 'New York', 0.15),

salesman_id	name	city	commission
5001	James Hoog	New York	0.15
5002	Nail Knite	Paris	0.13
5003	Lauson Hen	San Jose	0.12
5005	Pit Alex	London	0.11
5006	Mc Lyon	Paris	0.14
5007	Paul Adam	Rome	0.13
5008	Joen Rose	London	0.12
5009	Alex Smith	New York	0.11
5010	Sara Bond	Berlin	0.15
5011	Ben Lucas	Paris	0.14
5012	Kate Wood	London	0.12
5013	Rick Mort	Chicago	0.11
5014	Tina Fey	Rome	0.13
5015	Leo King	New York	0.12
5016	Mia Wong	San Jose	0.14
5017	Zoe Bell	London	0.15
5018	Sam Wise	Berlin	0.13
5019	Tom Hard	Paris	0.12
5020	Eva Long	Chicago	0.11
5021	Dan Rice	New York	0.14

20 rows in set (0.001 sec)

- INSERT INTO customer (customer_id, customer_name, city, grade, salesman_id) VALUES (3001, 'Brad Guzan', 'London', 100, 5005), etc etc...

customer_id	customer_name	city	grade	salesman_id
3001	Brad Guzan	London	100	5005
3002	Nick Rimando	New York	200	5001
3003	Jozy Altidor	Monzon	200	5007
3004	Fabian Johns	Paris	300	5006
3005	Graham Zusi	California	200	5002
3007	Brad Davis	New York	200	5001
3008	Julian Green	London	300	5002
3009	Geoff Cameron	Berlin	100	5010
3010	Tim Howard	Rome	400	5014
3011	Kyle Beck	Chicago	300	5013
3012	Deuce Dem	Paris	200	5011
3013	Mix Diske	London	300	5017
3014	Omar Gonz	New York	100	5015
3015	Matt Besl	San Jose	200	5016
3016	Alej Bedo	Berlin	400	5018
3017	Chris Won	Rome	200	5007
3018	Aron Joha	Chicago	300	5020
3019	John Broo	Paris	100	5019
3020	Timmy Cha	London	200	5005
3021	Nick DeLe	New York	500	5021

20 rows in set (0.001 sec)

- INSERT INTO orders (order_no, purchase_amt, order_date, customer_id, salesman_id) VALUES (70001, 150.50, '2023-10-05', 3005, 5002), etc etc...

```
MariaDB [labwork]> SELECT * FROM orders;
```

order_no	purchase_amt	order_date	customer_id	salesman_id
70001	150.50	2023-10-05	3005	5002
70002	65.26	2023-10-05	3002	5001
70003	2480.40	2023-10-10	3009	5010
70004	110.50	2023-10-17	3008	5002
70005	2400.60	2023-10-27	3007	5001
70007	948.50	2023-09-10	3005	5002
70008	5760.00	2023-09-10	3002	5001
70009	270.65	2023-09-10	3001	5005
70010	1983.43	2023-10-10	3004	5006
70011	75.29	2023-08-17	3003	5007
70012	250.45	2023-06-27	3008	5002
70013	3045.60	2023-04-25	3010	5014
70014	1200.00	2023-01-12	3011	5013
70015	850.75	2023-02-15	3012	5011
70016	450.00	2023-03-20	3013	5017
70017	1500.00	2023-04-10	3014	5015
70018	99.99	2023-05-18	3015	5016
70019	300.50	2023-06-22	3016	5018
70020	125.00	2023-07-30	3017	5007
70021	500.25	2023-08-12	3018	5020

20 rows in set (0.001 sec)

- Join them all together!

```
MariaDB [labwork]> SELECT
    ->     o.order_no,
    ->     c.customer_name,
    ->     o.purchase_amt,
    ->     s.name AS salesman_name
    -> FROM orders o
    -> JOIN customer c ON o.customer_id = c.customer_id
    -> JOIN salesman s ON o.salesman_id = s.salesman_id;
+----------+---------------+--------------+---------------+
| order_no | customer_name | purchase_amt | salesman_name |
+----------+---------------+--------------+---------------+
|    70001 | Graham Zusi   |       150.50 | Nail Knite    |
|    70002 | Nick Rimando  |        65.26 | James Hoog    |
|    70003 | Geoff Cameron |      2480.40 | Sara Bond     |
|    70004 | Julian Green  |       110.50 | Nail Knite    |
|    70005 | Brad Davis    |      2400.60 | James Hoog    |
|    70007 | Graham Zusi   |       948.50 | Nail Knite    |
|    70008 | Nick Rimando  |      5760.00 | James Hoog    |
|    70009 | Brad Guzan    |       270.65 | Pit Alex      |
|    70010 | Fabian Johns  |      1983.43 | Mc Lyon       |
|    70011 | Jozy Altidor  |        75.29 | Paul Adam     |
|    70012 | Julian Green  |       250.45 | Nail Knite    |
|    70013 | Tim Howard    |      3045.60 | Tina Fey      |
|    70014 | Kyle Beck     |      1200.00 | Rick Mort     |
|    70015 | Deuce Dem     |       850.75 | Ben Lucas     |
|    70016 | Mix Diske     |       450.00 | Zoe Bell      |
|    70017 | Omar Gonz     |      1500.00 | Leo King      |
|    70018 | Matt Besl     |        99.99 | Mia Wong      |
|    70019 | Alej Bedo     |       300.50 | Sam Wise      |
|    70020 | Chris Won     |       125.00 | Paul Adam     |
|    70021 | Aron Joha     |       500.25 | Eva Long      |
+----------+---------------+--------------+---------------+
20 rows in set (0.041 sec)
```

3. Create **DBConnection.java** that takes username/password/database name as commandline arguments and connects to the **labwork** database. The program should execute a query to retrieve data and create Sales objects for each order number and put them into a ArrayList<Sales>

```
import java.sql.*;
import java.util.ArrayList;

// Class to hold data
class Sales {
  int orderNumber;
  String customerName, customerCity, salesmanName;
  double amount, commissionAmount;

  // Constructor that does the math internally
  Sales(int id, String cn, String cc, String sn, double amt, double com) {
    orderNumber = id;
    customerName = cn;
    customerCity = cc;
    salesmanName = sn;
    amount = amt;
    commissionAmount = amt * com;
  }
}
```

```
public class DBConnection {
  public static void main(String[] args) throws Exception {

    String url = "jdbc:mariadb://localhost:3306/" + args[2];
    ArrayList<Sales> list = new ArrayList<>();

    // Join query to get all data at once
    String sql = "SELECT o.order_no, c.customer_name, c.city, s.name, o.purchase_amt, s.commission "
+
            "FROM orders o JOIN customer c ON o.customer_id=c.customer_id " +
            "JOIN salesman s ON o.salesman_id=s.salesman_id";

    try (Connection c = DriverManager.getConnection(url, args[0], args[1])) {
      ResultSet rs = c.createStatement().executeQuery(sql);
      while (rs.next())
        // Create Sales object and add to list
        list.add(new Sales(rs.getInt(1), rs.getString(2), rs.getString(3),
                    rs.getString(4), rs.getDouble(5), rs.getDouble(6)));
    } catch (Exception e) {
    e.printStackTrace();
  }

    // Print the objects from the list
    for (Sales s : list)
      System.out.println(s.orderNumber + " " + s.customerName + " $" + s.commissionAmount);
  }
}
```

```
Order 70001: Graham Zusi | Amount: $150.5 | Comm: $19.565
Order 70002: Nick Rimando | Amount: $65.26 | Comm: $9.789
Order 70003: Geoff Cameron | Amount: $2480.4 | Comm: $372.06
Order 70004: Julian Green | Amount: $110.5 | Comm: $14.365
Order 70005: Brad Davis | Amount: $2400.6 | Comm: $360.09
Order 70007: Graham Zusi | Amount: $948.5 | Comm: $123.305
Order 70008: Nick Rimando | Amount: $5760.0 | Comm: $864.0
Order 70009: Brad Guzan | Amount: $270.65 | Comm: $29.771499999999996
Order 70010: Fabian Johns | Amount: $1983.43 | Comm: $277.6802
Order 70011: Jozy Altidor | Amount: $75.29 | Comm: $9.787700000000001
Order 70012: Julian Green | Amount: $250.45 | Comm: $32.5585
Order 70013: Tim Howard | Amount: $3045.6 | Comm: $395.928
Order 70014: Kyle Beck | Amount: $1200.0 | Comm: $132.0
Order 70015: Deuce Dem | Amount: $850.75 | Comm: $119.10500000000002
Order 70016: Mix Diske | Amount: $450.0 | Comm: $67.5
Order 70017: Omar Gonz | Amount: $1500.0 | Comm: $180.0
Order 70018: Matt Besl | Amount: $99.99 | Comm: $13.998600000000001
Order 70019: Alej Bedo | Amount: $300.5 | Comm: $39.065000000000005
Order 70020: Chris Won | Amount: $125.0 | Comm: $16.25
Order 70021: Aron Joha | Amount: $500.25 | Comm: $55.0275
```